

Experiment 3: when does a push end, and why is a ramp not symmetric?

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Uploaded to DTU Learn by Anastasios Tatarakis for the group.

Dummy hand in for practice. All data are simulated.

Purpose

In Part 1 we test when a pushing hand loses contact with an iOLab that slides on a table: at the peak of the acceleration (A), when the acceleration is zero (B), or when the acceleration becomes constant (C). In Part 2 we compare the acceleration of the iOLab rolling up a ramp, a_1 , with the acceleration rolling down, a_2 , and explain the difference with the forces. We use the 3D accelerometer, the force probe and the wheel sensor of the iOLab.

1 Part 1: when does the hand lose contact?

1.1 Setup and method

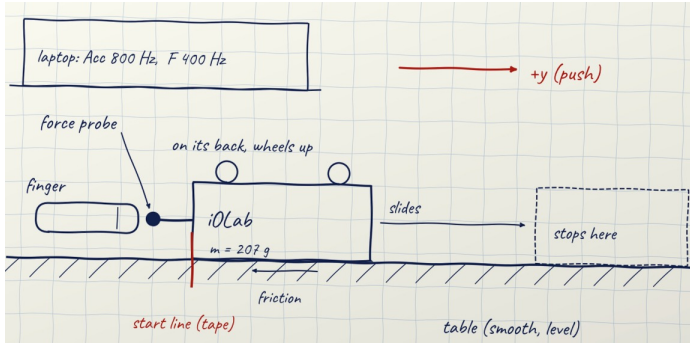


Figure 1. Sketch of the Part 1 setup, side view.

The iOLab, $m = (0.207 \pm 0.001)$ kg on the lab scale, lay on its back on a smooth, level table with its y axis along the push (Figure 1). We pushed it along $+y$ with a fingertip on the force probe, so the probe measured the whole force from the hand, F . After the push it slid and stopped by itself. The accelerometer (800 Hz) and the force probe (400 Hz) recorded together, with smoothing off, in 6 runs. Every recording starts with one quiet second, which sets the zero of both sensors and gives the force noise $\sigma_F = 0.010$ N. The push direction is positive.

1.2 Theory

While the hand pushes and the iOLab slides, Newton's second law along y gives

$$ma = F - \mu_k mg \quad \Rightarrow \quad F = m(a + \mu_k g). \quad (1)$$

When the hand has let go, $F = 0$ and only kinetic friction acts, so the acceleration is the constant

$$a = a_f = -\mu_k g. \quad (2)$$

By equation (1) the hand force is largest at A. At B, where $a = 0$ and the speed is largest, the hand still pushes with $F_B = \mu_k mg$. Only at C, where $a = a_f$, is $F = 0$, so the theory predicts C. A hand that only touches can push but not pull, so $F \geq 0$ and the acceleration cannot fall below a_f while the iOLab slides.

1.3 Hypothesis

Hypothesis C: the hand loses contact at point C, where the acceleration becomes constant. We find the moment the hand force is back at zero, t_{F0} (first sample inside $\pm 2\sigma_F$), read the acceleration there, $a(t_{F0})$, and compare it with the plateau a_f using the standard score $z = |x_1 - x_2| / \sqrt{\delta x_1^2 + \delta x_2^2}$. C is not rejected if $z \leq 2$ and rejected if $z > 3$. In the same way we compare $a(t_{F0})$ with 0 (B) and with the peak a_A (A).

1.4 Analysis

A is the maximum of a_y , B the first zero after A, and C the first sample after B inside $a_f \pm 2\sigma$. The plateau a_f is the mean over the middle of the slide, with uncertainty $\sigma/\sqrt{N}$ and $N = f\Delta t$. $a(t_{F0})$ is the mean of the samples within 2.5 ms of t_{F0} . Its uncertainty combines the noise, $\sigma/\sqrt{n}$, with the timing between the two sensors,

$\delta t = \sqrt{1/f_a^2 + 1/f_F^2} = 2.8$ ms, times the slope of the graph. Means over runs use the standard deviation of the mean (SDOM). The analysis was done in Python on the exported CSV files.

1.5 Results

Figure 2 shows run 5 and Table 1 all runs. Averaged over the 6 runs, the plateau is $a_f = (-1.878 \pm 0.018)$ m/s², so $\mu_k = -a_f/g = 0.1913 \pm 0.0018$. When the force is back at zero, $a(t_{F0}) = (-1.80 \pm 0.05)$ m/s². The peak is $a_A = (10.3 \pm 0.5)$ m/s². The contact ends $t_{F0} - t_B = (28.3 \pm 1.5)$ ms after B and $t_{F0} - t_C = (4.4 \pm 2.0)$ ms after C.

1.6 Discussion and conclusion

The difference $a(t_{F0}) - a_f = (0.08 \pm 0.05)$ m/s² gives $z_C = 1.4 \leq 2$: when the hand lets go, the acceleration has already reached the friction plateau. Hypothesis C is not rejected; only run 3 lands in the grey zone ($z = 2.1$). A and B are rejected: $a(t_{F0})$ differs from 0 with $z_B = 34$ and from the peak with $z_A = 22$. The force at B, $F(t_B) = (0.37 \pm 0.03)$ N, agrees with the friction force $\mu_k mg = (0.389 \pm 0.004)$ N ($z = 0.5$), so at B the hand still

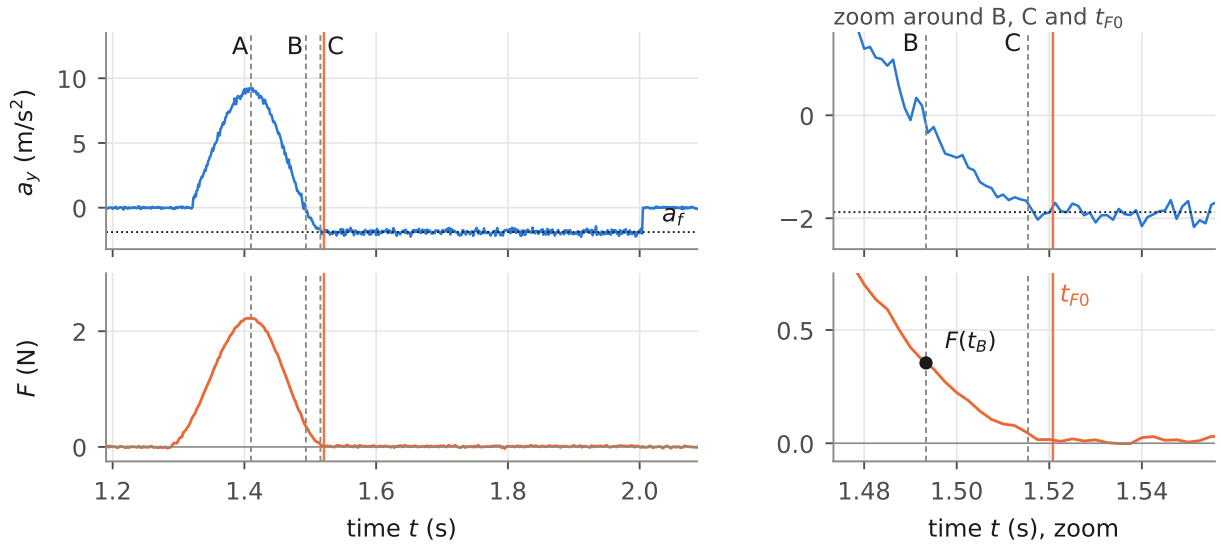


Figure 2. Part 1, run 5. Top: acceleration a_y . Bottom: hand force F from the force probe. Dashed lines mark A, B and C; the orange line marks t_{F0} . Right: zoom around B, C and t_{F0} . At B the hand still pushes, $F(t_B) = (0.35 \pm 0.06)$ N.

Table 1. Part 1, all runs. t_A , t_B , t_C are read to one accelerometer sample, ± 0.0013 s; t_{F0} to one force sample, ± 0.0025 s. z_C compares $a(t_{F0})$ with a_f ; z_B compares $a(t_{F0})$ with 0.

Run	t_A (s)	t_B (s)	t_C (s)	t_{F0} (s)	a_f (m/s ²)	$a(t_{F0})$ (m/s ²)	z_C	z_B
1	1.1275	1.2292	1.2570	1.2571	-1.825 ± 0.008	-1.7 ± 0.3	0.4	6.9
2	1.1775	1.2578	1.2729	1.2847	-1.892 ± 0.007	-1.84 ± 0.09	0.6	20.5
3	1.2375	1.3285	1.3559	1.3544	-1.828 ± 0.007	-1.64 ± 0.09	2.1	18.3
4	1.3550	1.4459	1.4688	1.4721	-1.921 ± 0.008	-1.97 ± 0.07	0.7	29.0
5	1.4100	1.4934	1.5154	1.5208	-1.881 ± 0.009	-1.84 ± 0.10	0.4	18.7
6	1.2188	1.3234	1.3516	1.3589	-1.922 ± 0.009	-1.79 ± 0.10	1.3	17.8
Mean					-1.878 ± 0.018	-1.80 ± 0.05	1.4	34

pushes exactly as hard as friction. This is what equation (1) predicts: $F = 0$ only when $a = -\mu_k g$. As a check of equation (1), a straight line fit of a against F during the push gives $m = (0.2020 \pm 0.0002)$ kg from the slope and an intercept of (-1.91 ± 0.02) m/s², which agrees with a_f ($z = 1.2$). The fitted mass is (2.4 ± 0.5) % below the scale value ($z = 4.8$); we think the force probe calibration, which we did not redo today, causes this.

2 Part 2: rolling up and down a ramp

2.1 Setup and method

A wooden board was raised at one end with books (Figure 3). Two tape marks $L = (1.000 \pm 0.002)$ m apart along the board had heights $h_1 = (2.1 \pm 0.1)$ cm and $h_2 = (9.1 \pm 0.1)$ cm above the table, so $\sin \theta = (h_2 - h_1)/L = 0.0700 \pm 0.0014$ and $\theta = (4.01 \pm 0.08)^\circ$. The iOLab stood on its wheels with y along the board and the measuring wheel on the board. Each run started at rest near the bottom. A short push sent the iOLab up the board; it turned, rolled back and was caught by hand. The wheel sensor recorded at 100 Hz. Up the board is positive. We made 6 runs from the same start mark.

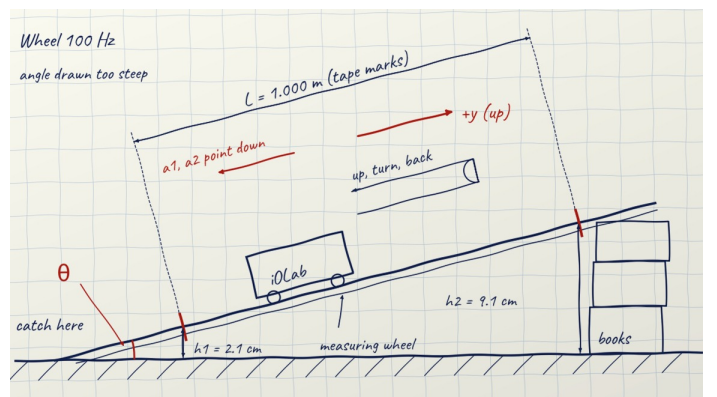


Figure 3. Sketch of the Part 2 setup, side view, angle drawn larger than real.

2.2 Theory and hypothesis

Along the board, gravity gives $-mg \sin \theta$ both ways. The rolling resistance F_r opposes the velocity, so it points down the board going up and up the board going down:

$$ma_1 = -mg \sin \theta - F_r, \quad ma_2 = -mg \sin \theta + F_r. \quad (3)$$

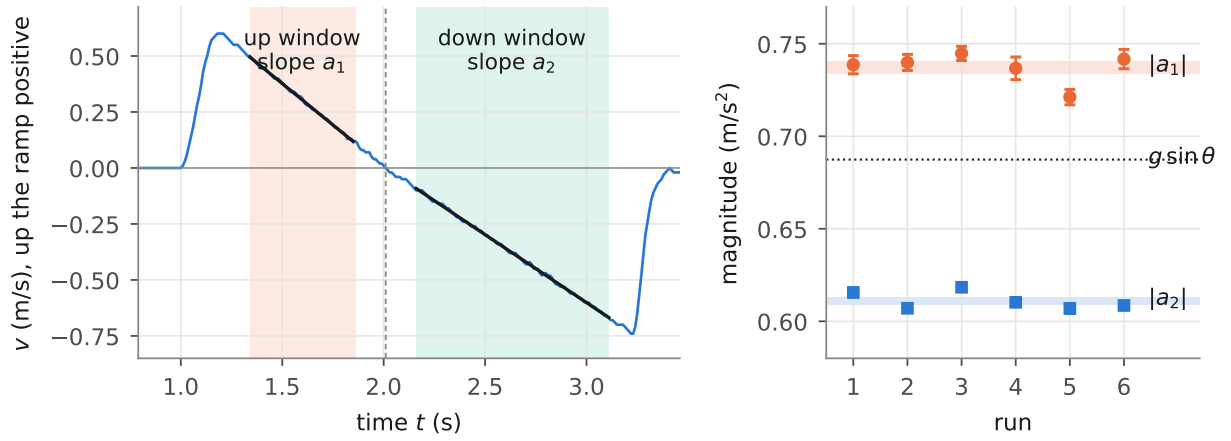


Figure 4. Part 2. Left: velocity of run 2 with the two fit windows; the slopes are a_1 and a_2 . Right: $|a_1|$ and $|a_2|$ for all runs with error bars; the bands show mean $\pm$ SDOM and the dotted line $g \sin \theta$ from the ruler.

Both accelerations point down the board, and $|a_1| = g \sin \theta + F_r/m$ is larger than $|a_2| = g \sin \theta - F_r/m$. Adding and subtracting gives

$$\frac{|a_1| + |a_2|}{2} = g \sin \theta, \quad \frac{|a_1| - |a_2|}{2} = \frac{F_r}{m}. \quad (4)$$

Hypothesis: $|a_1| > |a_2|$, because the rolling resistance adds to gravity going up and works against it going down. Test: $D = |\bar{a}_1| - |\bar{a}_2|$ with $\delta D = \sqrt{\delta \bar{a}_1^2 + \delta \bar{a}_2^2}$ and $z = D/\delta D$. The hypothesis is not rejected if $D > 0$ with $z > 3$, and rejected if $z \leq 2$ (then a_1 and a_2 are equal within the uncertainty).

2.3 Analysis and results

In each run we fitted a straight line to $v(t)$ in two windows (Figure 4, left). The up window runs from where the acceleration has settled after the push to 0.15 s before the turning point ($v = 0$). The down window runs from 0.15 s after the turning point to 0.10 s before the catch. The slopes are a_1 and a_2 , with the standard error of the fit as uncertainty (Table 2). As a check, the mean of the acceleration graph in the same windows, with uncertainty $\sigma/\sqrt{N}$, gives $a_1 = (-0.738 \pm 0.006) \text{ m/s}^2$ and $a_2 = (-0.613 \pm 0.005) \text{ m/s}^2$, the same within the uncertainty.

Table 2. Part 2, all runs. Slopes of $v(t)$ with the standard error of the fit. Last row: mean $\pm$ SDOM.

Run	$a_1 \text{ (m/s}^2\text{)}$	$a_2 \text{ (m/s}^2\text{)}$	$ a_1 - a_2 \text{ (m/s}^2\text{)}$
1	-0.739 ± 0.005	-0.616 ± 0.002	0.123 ± 0.006
2	-0.740 ± 0.004	-0.607 ± 0.002	0.133 ± 0.005
3	-0.745 ± 0.004	-0.6184 ± 0.0019	0.126 ± 0.004
4	-0.737 ± 0.006	-0.6103 ± 0.0019	0.126 ± 0.006
5	-0.721 ± 0.004	-0.607 ± 0.003	0.114 ± 0.005
6	-0.742 ± 0.005	-0.609 ± 0.002	0.133 ± 0.006
Mean	-0.737 ± 0.003	-0.6111 ± 0.0020	0.126 ± 0.004

2.4 Discussion and conclusion

$D = |\bar{a}_1| - |\bar{a}_2| = (0.126 \pm 0.004) \text{ m/s}^2$ gives $z = 32 > 3$, and every single run has $D > 0$. The hypothesis $|a_1| > |a_2|$ is not rejected: the iOLab slows down faster going up than it speeds up going down. The mean of the two, $(|\bar{a}_1| + |\bar{a}_2|)/2 = (0.6741 \pm 0.0020) \text{ m/s}^2$, agrees with $g \sin \theta = (0.687 \pm 0.014) \text{ m/s}^2$ from the ruler ($z = 0.9$), so the friction cancels as equation (4) predicts. The small shortfall fits the spinning wheels, which act like a slightly larger mass. Half the difference gives $F_r/m = (0.0630 \pm 0.0020) \text{ m/s}^2$, so $F_r = (0.0130 \pm 0.0004) \text{ N}$ and $\mu_r = F_r/(mg \cos \theta) = 0.00643 \pm 0.00020$. The rolling resistance opposes the velocity, so it changes direction at the turning point: going up it adds to the gravity component, going down it subtracts. We estimate air drag to be below 0.002 m/s^2 at these speeds, so it plays no role.

Error sources

Part 1: a time is only known to one sample (1.25 ms and 2.5 ms), which limits how well t_{F0} and t_C can be separated. Our C criterion ($a_f \pm 2\sigma$) marks C slightly early, which explains the positive $t_{F0} - t_C$. An offset of the force probe would make the force look nonzero after the contact; the quiet first second removes it. Part 2: if the push was not quite over at the start of the up window, $|a_1|$ comes out too small and hides the effect, so our D is, if anything, too small. The angle uncertainty only affects the $g \sin \theta$ check.

Reflections

1. How did you approach today's experiment: did you have expectations of the final result, or did you only get the result after you had the data?

Before the lab most of us expected B, the moment the iOLab is fastest. Newton's second law for the push pointed to C instead, so we built a test that could prove the theory wrong: the force probe gives the moment of contact loss independently of the accelerometer. For Part 2 we predicted $|a_1| > |a_2|$ from the free body diagrams before the first run; only the size of the difference came from the data.

2. What did you learn from today's experiment?

The hand keeps pushing after the moment of maximum speed; at B it still pushes as hard as friction. Reading one graph by eye would have given the wrong point, while a second sensor that measures the force settled it. On the ramp, friction changes direction at the top, so going up and coming down are not mirror images even though gravity is the same. We also learned to let the standard score decide.

3. What do you think you were supposed to learn from today's experiment?

To turn a question into a hypothesis that can fail, with a decision rule written down before the data. To design a measurement that can actually reject it, here by adding an independent sensor and repeating runs. And to explain the result with Newton's second law and free body diagrams, with uncertainties on every number, instead of trusting the shape of a graph.